

```
clc; clear; close all;
```

```
fp = 1000;  
fs = 8000;  
N = 50;
```

```
wc = fp/(fs/2);
```

```
b_hann = fir1(N, wc, 'high', hanning(N+1));  
disp('--- High-Pass Coefficients (Hanning Window) ---');
```

```
--- High-Pass Coefficients (Hanning Window) ---
```

```
disp(b_hann');
```

```
-0.0000  
 0  
 0.0003  
 0.0008  
 0.0009  
 0  
-0.0020  
-0.0038  
-0.0035  
 0  
 0.0057  
 0.0100  
 0.0087  
 0  
-0.0127  
-0.0216  
-0.0183  
 0  
 0.0267  
 0.0464  
 0.0410  
 0  
-0.0726  
-0.1568  
-0.2243  
 0.7500  
-0.2243  
-0.1568  
-0.0726  
 0  
 0.0410  
 0.0464  
 0.0267  
 0  
-0.0183  
-0.0216  
-0.0127  
 0  
 0.0087  
 0.0100  
 0.0057  
 0  
-0.0035
```

```
-0.0038  
-0.0020  
0  
0.0009  
0.0008  
0.0003  
0  
-0.0000
```

```
figure;  
freqz(b_hann,1);  
title('High-pass FIR using Hanning Window');
```

